

Architecture of a Software System for Protecting Wireless Ad Hoc Networks from Active Network Attacks

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Abstract—Software solutions for detecting and countering active network attacks on wireless ad hoc networks are analyzed. Requirements for the developed architecture of a software package for protecting wireless ad hoc networks from active network attacks are formulated. An architecture that meets the stated requirements is proposed. A software prototype implementing the proposed solution is developed and evaluated.

Keywords: information security, wireless ad hoc networks, network attack prevention, intrusion detection systems

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INTRODUCTION

The development of information technology contributes to the growth of the number of complex computer systems, whose purpose is to automate many aspects of human activity. An integral part of such systems are computer networks that unite disparate devices into a single system. Wireless ad hoc networks have a decentralized architecture and the ability to adapt to changes in network topology. Each node in these networks simultaneously plays the role of both an end device and a router. Due to their ability to adapt to changes in network composition, ease of deployment and scalability, ad hoc networks have found application in the implementation of smart city, transport, and industrial system concepts. They are also used in unmanned aircraft and can be used for rapid deployment in severe weather conditions and in the absence of network infrastructure.

However, the described advantages of wireless ad hoc networks have created new opportunities for attackers to conduct network attacks, and have also imposed certain limitations on systems protecting against network attacks. The potential use of ad hoc networks in critical information infrastructure confirms the need to develop systems to protect these networks from attacks.

ANALYSIS OF SOFTWARE SOLUTIONS AIMED AT DETECTING AND COUNTERING ACTIVE NETWORK ATTACKS IN WIRELESS AD HOC NETWORKS

Many researchers are studying the problem of protecting wireless ad hoc networks from network attacks. Among the proposed approaches, the following groups of solutions can be distinguished: cryptographic protection methods, solutions based on statistical analysis, signature methods, methods based on machine learning, and trust models.

In the work [1] an architecture of wireless ad hoc networks is proposed, based on the use of blockchain technology and software-defined networks (SDNs). The use of blockchain technologies ensures the confidentiality and integrity of data in the network, as well as distributed authentication of nodes in the network. SDN technology simplifies the routing process in the network. The proposed solution reduces node energy consumption and packet transmission delays, while it increases network throughput. The main disadvantage of the proposed solution is the violation of the decentralization property of the network due to the introduction of SDN controllers in the network. This feature opens a new attack vector for attackers: a successful attack on the SDN controller can lead to disruption of the network. The authors also do not introduce protection mechanisms against active attacks on the network by internal adversaries.

The source [2] proposes an architecture of a MANET network with an intrusion detection system based on a stack autoencoder (SAE) and a deep neural network (DNN). An SAE is used to reduce the features' dimensionality, while a DNN is used to classify attacks. This solution accurately detects active network attacks and can adapt to previously unknown attacks. This solution involves introducing additional nodes in the network, where network traffic will be collected and analyzed to detect attacks; therefore, the decentralization of the network is disrupted. The authors also do not propose mechanisms to counter malicious nodes, focusing solely on intrusion detection.

In [3], a routing protocol for mobile wireless networks based on the TBSMR trust model is proposed. This protocol is a modification of the AODV protocol, which improves the quality of service and security in the network, taking into account active network attacks. Malicious nodes are detected based on the statistical analysis of the operating parameters of nodes in the network. The advantages of the presented MANET network architecture are as follows: maintaining the property of decentralization, identifying active network attacks, and increasing network stability by using alternative routes and excluding nodes with low battery levels. The main disadvantages of the proposed solution are its limited accuracy of attack detection and its dependence on the attack method and its type.

The study [4] proposed an architecture based on a hybrid approach to ensure the security of MANET networks: cryptographic-based protection methods and a machine learning-based routing protocol were used. The FAES algorithm, which is a modification of the AES algorithm, is presented. The idea behind this algorithm is to use false encryption keys and mask data. FAES makes it more difficult for an attacker to eavesdrop on traffic and conduct certain types of active attacks. The EECBR routing protocol, whose main feature involves searching for routes based on the energy consumption data of nodes using the algorithm of the k -nearest neighbors, is also proposed. Malicious nodes are detected and excluded by periodic transmission of beacon requests to all nodes in the cluster. If the nodes do not respond to such requests, they are considered malicious [5]. The disadvantage of the proposed solution is that it only protects against a limited set of attacks. Moreover, when the approach to implementing attacks changes, the described solution loses its effectiveness.

In [6], an intrusion detection system based on the support vector machine (SVM) is presented. The proposed solution is aimed at detecting attacks at the network layer of the OSI model. Network activity is classified into two classes: normal and malicious. Although the proposed solution accurately detects attacks, it involves the implementation of privileged nodes and does not introduce attack countermeasure mechanisms.

The following main disadvantages were identified for the solutions considered: the introduction of privileged nodes, focus on a specific type of attack and the method of their implementation, and the lack of mechanisms to counter attacks.

REQUIREMENTS FOR THE PROPOSED ARCHITECTURE OF A SOFTWARE COMPLEX FOR PROTECTING WIRELESS AD HOC NETWORKS FROM ACTIVE NETWORK ATTACKS

As a result of the analysis of wireless ad hoc network protection systems in the previous section, the following requirements were formulated for the developed architecture of the software package for protecting wireless ad hoc networks:

1. Wide range of covered attacks: the proposed solution should be universal and provide protection against various types of active network attacks on ad hoc networks.
2. Accurate malicious activity detection: the number of type I and type II errors should be significantly lower than true positive and true negative alarms.
3. Adaptability: the proposed solution must be invariant to specific attack implementations.
4. High attack detection and response speed: the number of analyzed network packets related to the specified attack and required to detect it should be minimized.
5. Maintaining the decentralized nature of the network: the developed architecture should not contain privileged nodes.
6. Modularity: the proposed architecture of the system for protection against active network attacks on wireless ad hoc networks should consist of atomic, highly specialized modules, which ensure the scalability of the system and increase its fault tolerance.

ARCHITECTURE OF THE PROPOSED SOFTWARE PACKAGE

Taking into account the described requirements, an architecture of a system for protection against active network attacks on wireless ad hoc networks is proposed, presented in Fig. 1.

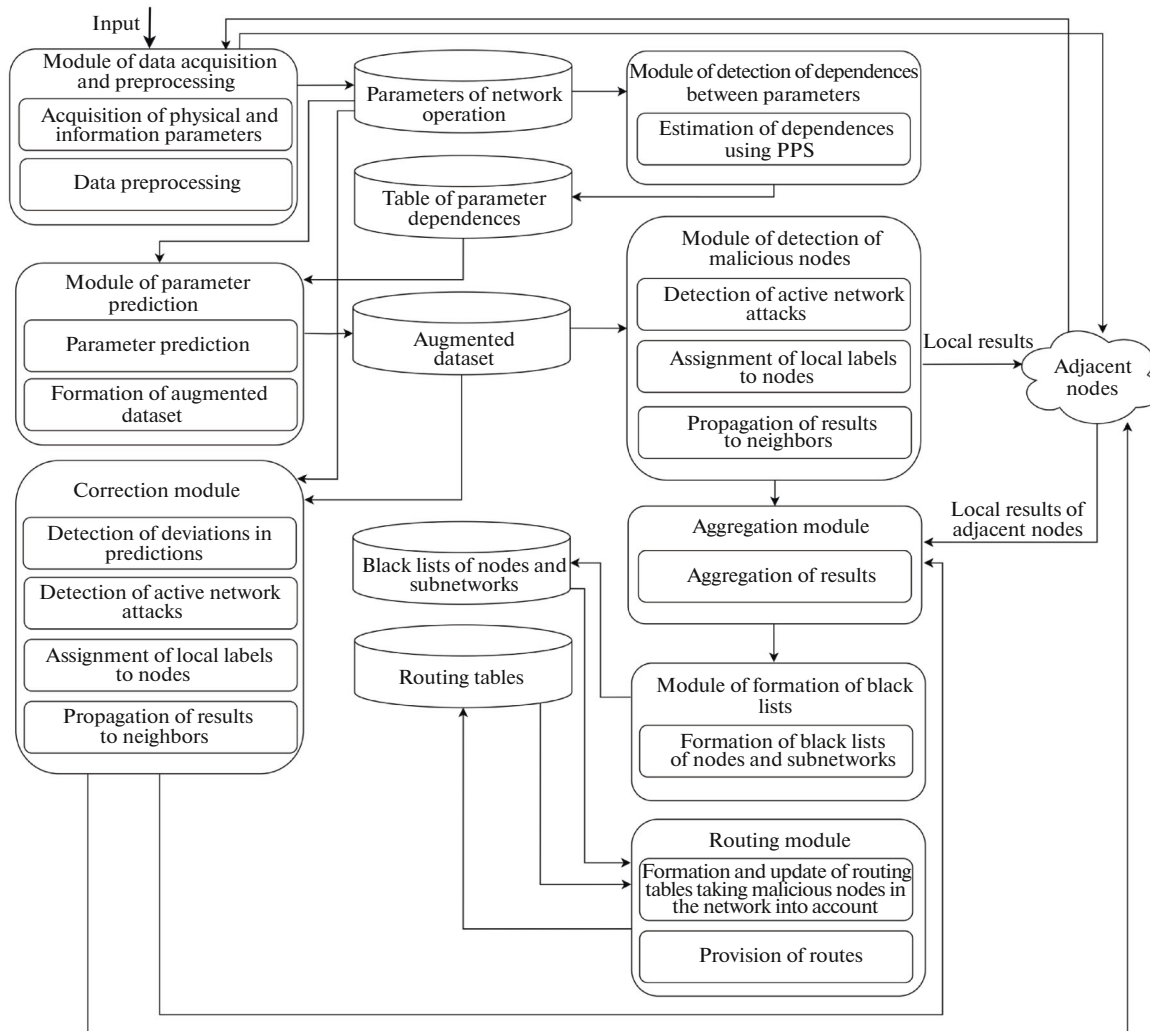


Fig. 1. Structural diagram of the architecture of the proposed solution.

The following main components are highlighted:

1. The data collection and preprocessing module collects characteristics of wireless ad hoc network nodes, extracts network traffic parameters, generates additional characteristics, normalizes the received data, and constructs a representation of network parameters in the form of time series.

2. The module for identifying dependencies between characteristics of data sets: for each parameter in the data set, it generates a set of related parameters, which are then used to forecast previously constructed time series.

3. The forecasting module is used to forecast the operating parameters of a wireless ad hoc network in order to increase the speed of detecting malicious activity in the network. The module is based on recurrent neural networks with long short-term memory.

4. Attack detection module: it analyzes the collected and predicted data to detect network attacks and identify malicious nodes. Implemented based on the SVM.

5. The response module is the module responsible for disseminating information about malicious nodes, creating and updating blacklists of malicious nodes and subnets, and excluding malicious nodes from the network.

6. The correction module reanalyzes the data corresponding to a certain time interval in order to correct errors that arose during the analysis of the combined set of collected and predicted data.

The described system operates on each individual node, which eliminates the need to introduce privileged nodes and does not violate the decentralization properties of the network. In this case, each node analyzes only its own parameters and the parameters of neighboring nodes rather than the parameters of the entire network, which significantly reduces the computational load.

Data collection and preprocessing module. This module operates on each node of the wireless ad hoc network and collects physical and informational parameters of the current node, as well as parameters of neighboring nodes.

The following physical parameters of nodes are identified: the position of the node in space at the given moment in time t , presented as a tuple of coordinates and altitude above sea level; the node velocity (km/h) at time t ; and the range of wireless communication modules in meters.

The following node parameters are classified as informational:

- node identifier;
- mobility flag, which takes value 1 if the node is mobile, and 0 otherwise;
- infrastructure membership flag, which takes value 1 if the node is part of the network deployment infrastructure (for example, a base station, roadside infrastructure, etc.), and 0 otherwise;
- the node's trust level, which takes a value from 0 to 1.

This module also collects network traffic. This module extracts the following parameters from the network traffic collected by a node: source node identifier, destination node identifier, timestamp, packet size in bytes, packet sequence number, TCP/IP application/transport layer protocol used, packet flags.

The following characteristics are additionally calculated from the collected data: number of packets transmitted; number of packets received; average number of packets transmitted per unit of time; average number of packets received per unit of time; ratio of packets transmitted to those received; average size of message transmitted; and average size of message received.

After collecting the data, extracting the basic characteristics from them, and calculating additional characteristics, they are normalized and transferred to the prediction module and the detection module.

Module for identifying dependencies between characteristics of data sets. The dependency detection module is used to evaluate dependencies between the described parameters. To evaluate dependencies, the predictive power score (PPS) algorithm is used. Through it, a set of related attributes is formed for each attribute. The information obtained is used at the forecasting stage to build more accurate forecasts and improve the performance of the models.

This module, unlike others, functions only during preliminary configuration and system training.

Forecasting module. This is an ensemble of recurrent neural networks with long short-term memory. In this module, forecasting of the described parameters is carried out with the given forecasting step.

Each parameter is predicted based on the previous values of that parameter, as well as on the values of related attributes obtained by the dependency detection module. To predict the parameters, a Seq2Seq model with an attention mechanism, built based on recurrent neural networks with long short-term memory, is used. The architecture of the Seq2Seq model consists of an encoder that transforms a multidimensional input sequence into a context vector, and a decoder that generates data based on the context vector and previously predicted values [7]. The classical Seq2Seq model has the drawback that each decoder step depends on the same vector received by the encoder. The attention mechanism allows the creation of a context vector specific to each step of the decoder; i.e., the decoder focuses on different parts of the input vector, which allows for increased prediction accuracy.

The Seq2Seq model allows accurately forecasting values for multiple steps; however, in order to reduce the computational load on nodes, a decision was made to use a hybrid forecasting approach that combines multistep and iterative forecasting. Figure 2 shows a diagram for predicting a certain parameter of an ad hoc network.

Active network attack detection module. This module combines collected and predicted data for further analysis. The purpose of the analysis is to divide nodes into two classes: malicious and legitimate [8]. The SVM was chosen to solve this problem.

The application of machine learning methods to network attack detection tasks makes it possible to identify a wide variety of attacks, regardless of their implementation method. According to the works [9–12], the SVM method detects network attacks on wireless ad hoc networks more accurately than other machine learning-based methods. The analysis results are distributed to neighboring nodes.

Response module. This ensures that malicious nodes are excluded from the network. For each node, a confidence level is calculated based on the values obtained as a result of the data analysis by the current node and the values obtained by neighboring nodes. If the resulting trust level is lower than some threshold

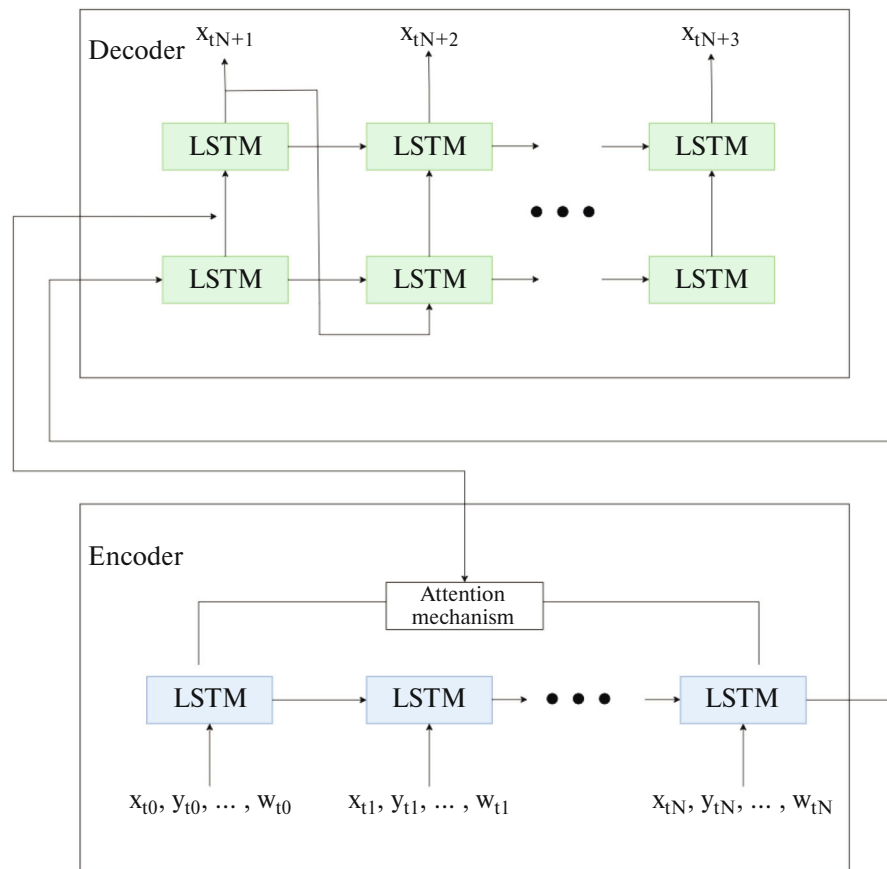


Fig. 2. Schematic diagram of the forecasting module.

value, the node in question is added to the blacklist, which triggers the process of updating the routing tables. Nodes added to the blacklist are excluded from the network.

To increase the speed of response to groups of external malicious nodes, a submodule for compiling blacklists of subnets was implemented [13].

Correction module. This is used to adjust the results of identifying malicious nodes under the condition that the operating parameters of the wireless ad hoc network obtained using the forecasting module differ significantly from the parameters obtained by the collection module over a similar period of time.

The correction module duplicates the active network attack detection module, except that this module receives as input a data set obtained only by the collection module.

EVALUATION OF THE PROPOSED SOLUTION

Publicly available datasets describing the performance of ad hoc networks under various network attacks do not contain all the parameters required to evaluate the proposed solution. Therefore, datasets collected from NS3 [14] simulations of wireless ad hoc networks under the influence of the following active network attacks were used: DoS, DDoS, Sybil, Blackhole, and Grayhole.

The accuracy and speed of detecting active network attacks using the SVM model were evaluated for two scenarios: one in which only the collected network operating parameters were used as the input data; and the other in which the collected network operating parameters and the parameters obtained using the forecasting module were used as input data.

In predicting the network performance parameters, the mean absolute percentage error (MAPE) was 8%. The results of the accuracy and speed of detection of active network attacks are presented in the Table 1.

Table 1. Results of the evaluation of the proposed solution

Scenario index	Accuracy	Precision	Recall	Detection speed, s
DDoS				
1	0.97	0.96	0.98	86
2	0.97	0.95	0.98	21
DoS				
1	0.98	0.94	0.98	73
2	0.97	0.91	0.97	17
Sybil				
1	0.96	0.96	0.95	741
2	0.94	0.92	0.95	428
Blackhole				
1	0.97	0.98	0.92	383
2	0.97	0.97	0.92	195
Grayhole				
1	0.96	0.94	0.99	1008
2	0.95	0.92	0.99	694

The experimental results demonstrate that the proposed solution, with a slight decrease in accuracy, allows for a reduction in the time spent on detecting active network attacks on average by factors of 2–3.

To evaluate the subnet blacklisting module, a dataset was used that included lists of malicious and legitimate nodes on the Internet IPsum [15] and Dataset of Malicious and Benign Webpages [16]. The compiled dataset contains approximately 715000 addresses, of which 162000 records are malicious nodes and 553000 records are legitimate. In the experiment, 50000 records were randomly extracted from the compiled data set and fed to the input of the module. For a threshold corresponding to a /24 subnet mask, the proportion of previously unseen malicious nodes blocked was approximately 10%, and the proportion of legitimate nodes blocked was approximately 0.008%.

CONCLUSIONS

The architectures of systems for protecting wireless ad hoc networks from network attacks are analyzed, and their advantages and disadvantages are highlighted.

Based on the analysis, the following requirements were formulated for the developed architecture of the system for protection against active network attacks on wireless ad hoc networks: a wide range of covered attacks, accurate detection of malicious activity, adaptability, high speed of detection and response to attacks, preservation of the network's decentralization properties, and modularity.

An architecture of a system for protection against active network attacks on wireless ad hoc networks is proposed, following the formulated requirements. The software implementation of the proposed solution was completed and its evaluation was carried out. The experimental results demonstrate that the proposed solution can reduce the detection time of active network attacks on average by factors of 2–3, depending on the type of attack and the method of its implementation. At the same time, the accuracy of attack detection remains at a high level. The proposed solution also allows faster blocking of attacker nodes carrying out a distributed attack on the network.

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CONFLICT OF INTEREST

The author of this work declares that he has no conflicts of interest.

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